An AI-Driven Gamified Skill-Tree System for Self-Directed Learning

Student Name: Tianhao Li
Supervisor Name: Yang Long
Submitted as part of the degree of BSc Computer Science to the
Board of Examiners in the Department of Computer Sciences, Durham University

Abstract—This report presents a literature-grounded design for an AI-driven gamified skill-tree learning system and a three-tier project implementation plan. The core objective is to reduce cognitive overload in self-directed learning, improve sustained motivation through gamification, and provide dynamic scaffolding with large language models. The proposed system models learning dependencies as a directed acyclic graph and supports personalized path optimization. The implementation roadmap covers foundational infrastructure, personalization via retrieval-augmented generation, and advanced quantitative evaluation with gamified feedback mechanisms.

Index Terms—Micro-learning, Gamification, Self-regulated learning, Large language models, Knowledge graph, Personalized learning

1 Part 1: Literature Review

1.1 Cognitive Foundations: Reducing Overload through Micro-learning

The primary challenge for self-directed learners in the digital age is information explosion, which often leads to cognitive paralysis. According to Cognitive Load Theory (CLT), proposed by Sweller (1988), human working memory has limited capacity [5]. When learners are presented with complex and unstructured goals, extraneous cognitive load can exceed processing ability and reduce learning efficiency.

Hug (2005) argues that micro-learning, which fragments knowledge into small and digestible units, is an effective response to this overload [2]. By decomposing a macro-objective into a series of nodes on a skill tree, the proposed system keeps each learning task within the learner's germane cognitive capacity. Traditional micro-learning platforms provide fragmented content, but often lack the structural coherence needed for deep mastery. This project addresses that gap through a structured skill-tree hierarchy.

1.2 Motivational Drivers: Gamification and Self-Regulated Learning

Gamification is defined by Deterding et al. (2011) as the use of game design elements in non-game contexts [1]. In education, the lighting-up mechanism of a skill tree functions as a strong visual feedback signal.

Building on Self-Determination Theory (Deci & Ryan, 2000), progression through a skill tree supports the learner's need for competence. Zimmerman (2002) emphasizes that self-regulated learning depends heavily on timely feedback loops [7]. Unlike traditional textbooks, immediate visual rewards from unlocking new nodes can sustain long-term engagement. Existing platforms such as Duolingo achieve strong engagement but can prioritize streak mechanics over knowledge dependency depth. This project combines high engagement gamification with rigorous curriculum-level logical constraints, consistent with motivation-centered learning design principles [1], [7].

1.3 Scaffolding Theory: Dynamic ZPD in the Age of AI

Vygotsky's Zone of Proximal Development (ZPD) states that optimal learning happens when tasks are slightly beyond independent ability but achievable with guidance [6]. Traditional personalized scaffolding is labor-intensive. Kasneci et al. (2023) identify large language models as a paradigm shift that can provide real-time personalized tutoring [4].

The proposed system transforms static scaffolding into dynamic scaffolding. By assessing user progress and generating the next logical step in the skill tree, the system keeps learners within their ZPD and avoids both boredom (tasks too easy) and anxiety (tasks too hard).

1.4 Technical Mapping: Knowledge Graphs and Path Optimization

From a structural perspective, a skill tree is a directed acyclic graph (DAG). Knowledge graph construction research (Ji et al., 2021) demonstrates how relationships among concepts can be represented as nodes and edges [3].

Traditional personalized learning path recommendation systems rely on predefined rule-based traversal over static graphs and struggle with rapidly evolving domains. Integrating large language models enables automated curriculum generation where the graph can be constructed or adapted in real time according to user goals. This addresses the rigidity of expert-curated fixed content and combines the mathematical rigor of DAGs with generative AI flexibility for lifelong learning.

2 Part 2: Project Plan

2.1 Three-Tier Development Strategy

To ensure robust architecture and implementation feasibility, development is divided into three tiers.

2.1.1 Tier 1: Foundational Logic and Core Infrastructure

Focus: establish MVP capabilities and data persistence.

AI Path Topology: engineer prompts for stable JSON skill-tree generation with DAG logical consistency.

System Infrastructure: implement secure login and registration (JWT), and design database schema for user metadata, unlocked nodes, and prerequisite states.

Outcome: users can log in, input a learning goal, and view a personalized prerequisite-locked learning tree.

2.1.2 Tier 2: Engineering Enhancement and Knowledge Management

Focus: personalization through retrieval-augmented generation (RAG) and interactive learning management.

RAG Pipeline: integrate vector retrieval over user learning history for context-aware mentoring.

Knowledge Persistence: implement state persistence so the system remembers prior queries and quiz outcomes.

Outcome: an intelligent platform that provides continuous personalized tutoring with a structured management dashboard.

2.1.3 Tier 3: Advanced Evaluation and Gamified Ecosystem

Focus: quantitative assessment and immersive feedback.

Algorithmic Evaluation: adopt Zhang-Shasha Tree Edit Distance to quantify similarity between user learning paths and expert benchmarks.

RPG Integration: implement automated experience points and leveling based on node completion and similarity scores, with visual node-lighting and badge feedback.

Outcome: a high-fidelity gamified learning system with objective mathematical feedback on learning professionality.

3 Part 3: Project Timeline (Gantt-Style)

Period	Phase	Milestone
27 Apr – 1 May 2026	Phase A: Kick-off & compliance	Literature review and project plan submitted; ethical/risk requirement status confirmed.
2 May – 16 May 2026	Phase B: Core MVP	Authentication, learner-state persistence, DAG skill-tree generation, and prerequisite unlocking completed.
17 May – 30 May 2026	Phase C: Personalization	RAG-based adaptation and persistent tutoring memory integrated; preliminary demonstration delivered.
31 May – 26 Jun 2026	Phase D: Evaluation & gamification	Path-similarity evaluation, EXP/level logic, and visual progression feedback implemented.
July 2026 – early Aug 2026	Phase E1: Dissertation drafting	Preliminary report chapters delivered to supervisor and feedback meeting completed.
Early Aug 2026 – 4 Sep 2026	Phase E2: Finalization & submission	Final report, code package, and project log completed and submitted by 4 Sep 2026 (2:00 PM).

TABLE 1
Project milestones and schedule in Gantt-style form.

4 References

References

[1] S. Deterding, D. Dixon, R. Khaled, and L. Nacke, "From game design elements to gamefulness: defining gamification," in *Proceedings of the 15th International Academic MindTrek Conference*, 2011, pp. 9–15, doi: https://doi.org/10.1145/2181037.2181040.

[2] T. Hug, "Micro learning: A new pedagogical challenge (introductory note)," in *Proceedings of Microlearning Conference 2005: Learning and Working in New Media*, Innsbruck University Press, 2005. [Online]. Available: https://www.researchgate.net/publication/272178142_Microlearning_A_New_Pedagogical_Challenge

[3] S. Ji, S. Pan, E. Cambria, P. Marttinen, and P. S. Yu, "A survey on knowledge graphs: Representation, acquisition, and applications," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 33, no. 2, pp. 494–514, 2022, doi: https://doi.org/10.1109/TNNLS.2021.3070843.

[4] E. Kasneci *et al.*, "ChatGPT for good? On opportunities and challenges of large language models for education," *Learning and Individual Differences*, vol. 103, p. 102274, 2023, doi: https://doi.org/10.1016/j.lindif.2023.102274.

[5] J. Sweller, "Cognitive load during problem solving: Effects on learning," *Cognitive Science*, vol. 12, no. 2, pp. 257–285, 1988, doi: https://doi.org/10.1207/s15516709cog1202_4.

[6] L. S. Vygotsky, *Mind in Society: The Development of Higher Psychological Processes*. Cambridge, MA: Harvard University Press, 1978. [Online]. Available: https://www.hup.harvard.edu/books/9780674576292

[7] B. J. Zimmerman, "Becoming a self-regulated learner: An overview," *Theory Into Practice*, vol. 41, no. 2, pp. 64–70, 2002, doi: https://doi.org/10.1207/s15430421tip4102_2.